Authorization

Trust the caller context. Recheck authority on replay.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A venue badge checked at each controlled door: a past entry receipt does not restore permission after the badge is revoked.

Why it matters: Replay, cached receipts and provider selection must not bypass current scope, method, route or approval policy. Identity must come from trusted authentication.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

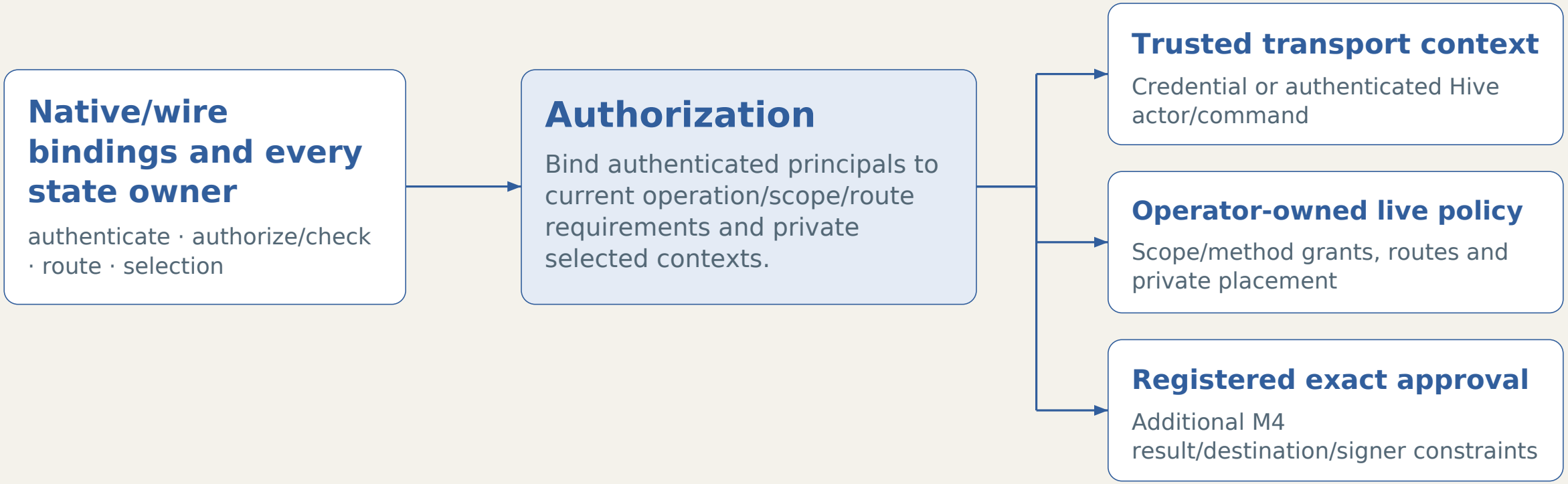
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	**Authorization**

Bind authenticated principals to current operation/scope/route requirements and private selected contexts.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Current policy configuration

Credentials, grants, routes, origin mappings and selected contexts

No business-effect journal

Each domain owner checks authority around its own effects and replay

OUTSIDE THIS RESPONSIBILITY

Caller-selected identity

No sender field, scope name or parameter bundle can grant itself authority

Production identity system

No claim of remote attestation, secrecy from administrators or full IAM

local-test-policy/1; private authority/selection ports; exact M4 approval constraints

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
authenticate	Validate credential; derive registered principal	Trusted transport, not user body identity
check / authorize	Require current scope and operation grant	Recheck cached replay and sensitive owner commits
route	Validate scope and route owner	Only owner polls/acks; worker route checked
Private selection / approval	Bind profile, placement and exact eligible result	Descriptor consistency is separate from permission

Trusted local host/operator; directory placement is not encryption or administrator secrecy.

A worked example through the boundary

REAL Hive authentication · explicitly local operator policy and generated grants

01 · STEP

Authenticate

Hive supplies actor and command from native context.

02 · STEP

Bind privately

Operator mapping selects scope, credential and provider.

03 · STEP

Revoke

Policy changes while an intent is pending.

04 · STEP

Replay refuses

Saved success does not bypass current authority.

What the example establishes

A valid old receipt can still require a current denial.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Trusted identity source

The request body never overrides authenticated principal.

I2

Current authority

Replay is not an authorization cache.

I3

Least owner access

Scope/method/route checks remain at declared boundaries.

I4

Private-by-placement

Public receipts/log parameters avoid private values and identifying hashes.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Worker permission revoked before restart	Recheck at work/recovery; no unauthorized result success.	Legacy worker-revocation tests
Route revoked after job completion	Retain outbox until authorized delivery can proceed.	No dropped completion
Private/chain approval changed before replay	Reject altered selection or ineligible intent.	Current checks and exact-result refusal tests

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

4

Worker-revocation matrix cases

Two storage providers × two bindings, historical focused recheck

LIVE

Policy read on current checks

Implementation behavior; no cached grant

NO SLO

Production authorization benchmark

Security properties require adversarial validation, not only speed

Current evidence is refusal, revocation, canary/privacy and trusted-identity testing. No production attack audit, authorization latency distribution or remote security certification exists.

PROPOSED BENCHMARK · NOT RUN

Proposed: measure authenticate/check/route p50/p95 across policy size and revocation churn; pair every timing run with unauthorized/replay/identity-spoof refusal checks.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Authenticated native Hive dispatch and executed current-policy checks in local owners.
FIXTURE / LOCAL TEST POLICY	Local test credentials/grants/routes, operator mappings, canaries and malformed requests.
BLOCKED / UNSUPPORTED	No required supported gate blocked; absent grants or revoked access are deliberate denials.
UNRUN	Production IAM, adversarial host compromise, remote attestation, distributed revocation and security audit.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Harden identity integration	Explicit production threat model and trust roots	No body-field impersonation or stale-grant replay
Measure revocation behavior	Controlled churn with concurrent reads/retries	Denial timing and false-acceptance checks together
Add audit views carefully	Authorized redacted event policy	Explain denials without leaking private data or credentials

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

A valid old receipt can still require a current denial.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m5.py --environment <env> --chain-build <chain> --output <fresh>
```

The complete supported regression covers current authority across legacy and new lookup paths. Native authentication and local policy remain separately labelled.

`fusion/policy.py`
`fusion/private.py`
`fusion/service.py`
`tests/test_m11.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.